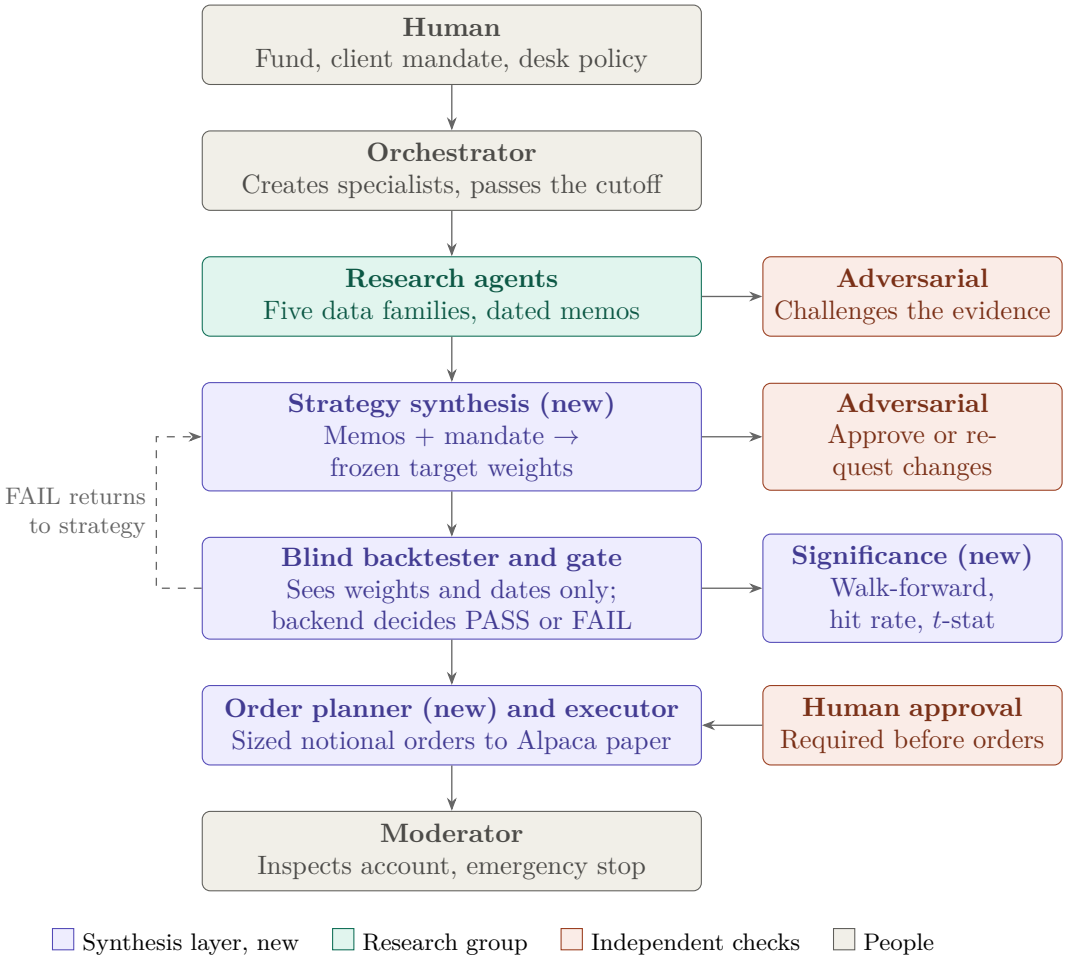
# Agent flow and variables

Team 4, Track 3: Financial Portfolio Management

Austin Agentic AI Hackathon, 23 October 2026

Two views of the trading system as it exists on `main`, plus the synthesis layer added in `backend/app/synthesis/`. Roles and tools come from `backend/app/roles.py`, gates from `backend/app/desk.py`, execution from `backend/app/paper.py`. Boxes marked **new** are the synthesis-layer additions.

## 1. High-level flow




**Gates the model cannot bypass.** A strategy must be registered (frozen) before review. The backtester is created only after an approving review and sees nothing but the weights and dates. The backend computes PASS or FAIL from human thresholds. Execution rechecks permission, PASS, review, approval, limits and the stop before every order. A submitted order is not a fill.

## 2. Which variables each agent uses

| Agent | Reads | Produces | Bound by |
|---|---|---|---|
| Research, live | Financial Datasets statements, ratios, earnings, prices, filing sections, insider and institutional ownership, company news; Exa web search; X search; FRED series; Jev sentiment; Apify | One memo per data family: ticker, stance (long, neutral, avoid), confidence 0 to 1, note, source ids with dates | Live data never enters a historical decision |
| Research, historical | Cutoff-filtered snapshots: OHLCV closes; SEC revenue, net income, total debt keyed on filing date; FRED CPI, FEDFUNDS, DGS10, UNRATE keyed on release date; curated events and GDELT tone | Same memo shape; every source dated on or before the cutoff | `assert_no_lookahead` on every cited source |
| **Strategy synthesis** (new) | All memos; mandate universe, capital, risk band, max_position, max_drawdown; policy universe and max_position | Frozen strategy: name, weights summing to 1, cutoff, end, rationale, memo paths, changes_my_mind. Seed is 1 plus signed convictions; net-avoid names dropped; clamped to max_position; optional single Bedrock refinement that is re-clamped in code | `register_strategy` rejects weights outside the universe or above max_position; revisions get new ids |
| Adversarial | The research or the registered strategy, plus native evidence tools | APPROVE or CHANGES_REQUESTED with a critique | Cannot review a strategy by its own author |
| Backtester | Only strategy id, weights, cutoff, end. The engine adds closes, SPY and the risk-free rate (FEDFUNDS at the cutoff) | total_return, cagr, max_drawdown, sharpe for portfolio and SPY | No file, web, research or delegation tools; bound to one strategy id |
| Gate | The metrics above; policy max_drawdown, min_sharpe, min_excess_return | PASS or FAIL with per-check results and a policy hash | Backend computes the verdict; a changed policy invalidates PASS |
| **Significance** (new) | Frozen weights; every monthly cutoff in range; fixed horizon | Number of windows, hit rate vs SPY and equal weight, mean excess return, $t$-statistic, $p$-value, bootstrap 95% interval | Reported beside the gate, never replaces it; windows overlap, so the $t$-stat is optimistic |
| **Order planner** (new) | Weights; capital as the smaller of account equity and mandate capital; account cash, positions, open orders; max_order_notional, max_daily_orders | Ordered list of buy and sell notional orders with stable client_order_ids, or blocked with a reason | Sells before buys; nothing above cash, position cap or notional limit |

| Agent | Reads | Produces | Bound by |
|---|---|---|---|
| Executor | The order plan; live web context only as a reason to defer | Broker order id, status, filled_qty, filled_avg_price | Stop clear, trading_allowed, current PASS, APPROVE, human approval and limits rechecked per order; an unknown result is never resubmitted |
| Moderator | Desk records, paper account, positions versus submitted orders | `emergency_stop` with a reason | Can engage the stop; only a person clears it |

## 3. How this maps onto the roles on `main`

Kushagra's written role requirements take priority over the picture: the roster is flexible, one root can create more orchestrators and task-focused researchers, and the boxes above are context, not a fixed roster. The synthesis boxes are **tools on existing roles**, not new roles: `synthesize_strategy` on the strategy role; `plan_paper_orders` and `execute_order_plan` on the executor role, which submits a plan one order at a time through the guarded adapter, halts at the first blocked or uncertain order, and resumes after `alpaca_paper_order` confirms the open order's real fill.

**Two evaluations, one contract.** Synthesis hands every evaluator the same frozen strategy record (weights, cutoff, end) by `strategy_id`. The desk gate on `main` runs the monthly-rebalance snapshot engine to 2025-06-30 and decides PASS or FAIL against the human policy. Mayank's walk-forward harness (D21, D22, accepted) is the leakage-safe evaluation shown beside it: weekly decisions from 2025-10-03 on Claude Opus 4.6, whose documented cutoff precedes the window, with strict point-in-time data and a memorization audit. Its registered adapters `synthesis_seed` and `synthesis_planner` wrap the synthesis seed and order planner, and its conformance suite proves they are pure functions of the point-in-time view. Reported honestly, his frozen run shows the agent pipeline trailing equal weight over 47 clean weeks; the method is the strength. The significance report in section 2 stays as an allocation-robustness add-on.

## 4. Limits to state in the demo

- Historical evaluation is the recorded monthly target-weight engine on cached closes: no dividends, no costs, fully invested. Significance is a walk-forward over cutoffs with the same frozen weights, which measures robustness of the allocation, not of a live rebalancing process.
- Live provider data is not point-in-time safe and never feeds a historical decision.
- Hiding research from the evaluator does not remove model-memory leakage; see Mayank's `docs/track3-plan.md` for the masked-ticker audit.
- Alpaca paper is a shared account; a submitted order is not a fill.